Using GIS in R code to Explore Links Between Smoking, Population Density, and Lung Cancer

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Abstract:

Cigarette and tobacco consumption has long been established as a high contributing factor towards the overall lung cancer incidents and other respiratory diseases. To assess the relationship between tobacco use and lung cancer incidence, three GIS models were constructed using data on lung cancer cases, tobacco prevalence, and the population distribution. The resulting analyses demonstrated a positive correlation, with regions that have high smoking rates reporting greater disease cases. (See Figure 1) Areas with high population density also tend to have higher lung cancer rates, indicating a variety of factors that contribute to increased lung disease. (See Figure 2 & 4) These findings highlight the importance of integrating public health data with spatial analysis to better understand global disease patterns.

Background Information:

- Lung cancer resulting from smoking tobacco products accounts for 90% of all lung cancer incidents which carries high fatality rates
- Inhaling tobacco smoke releases thousands of chemicals into the lungs, many of which alter and damage the DNA of the lungs
- The accumulation of damage in the lungs may eventually cause the formation of cancer cells.
- The chemicals can damage the alveoli, obstructing their ability to perform gas exchange
- People suffering from second hand smoking may also be at risk of lung cancer, though the percentage is drastically lower than smokers.

Method/Materials

- GIS maps were created through R studio, using the packages ggplot2, rnaturalearth, sf, countrycode, readr, tidyr, scales, and ggspatial.
- Lung cancer data was obtained from the WHO Global Cancer Observatory, and tobacco usage and smoking prevalence data was obtained from the World Health Organization
- Country names standardized using country code, and incomplete data was dropped, while the sf was used for mapping.
- Continuous color scale was used for the three maps, with plasma for population, magma for lung cancer, and cividis for smoking prevalence.

Results:

There appears to be a positive correlation between tobacco prevalence and lung cancer rates. Additionally, areas with high population densities appear to have a higher tendency to exhibit higher tobacco prevalence and lung cancer rates.

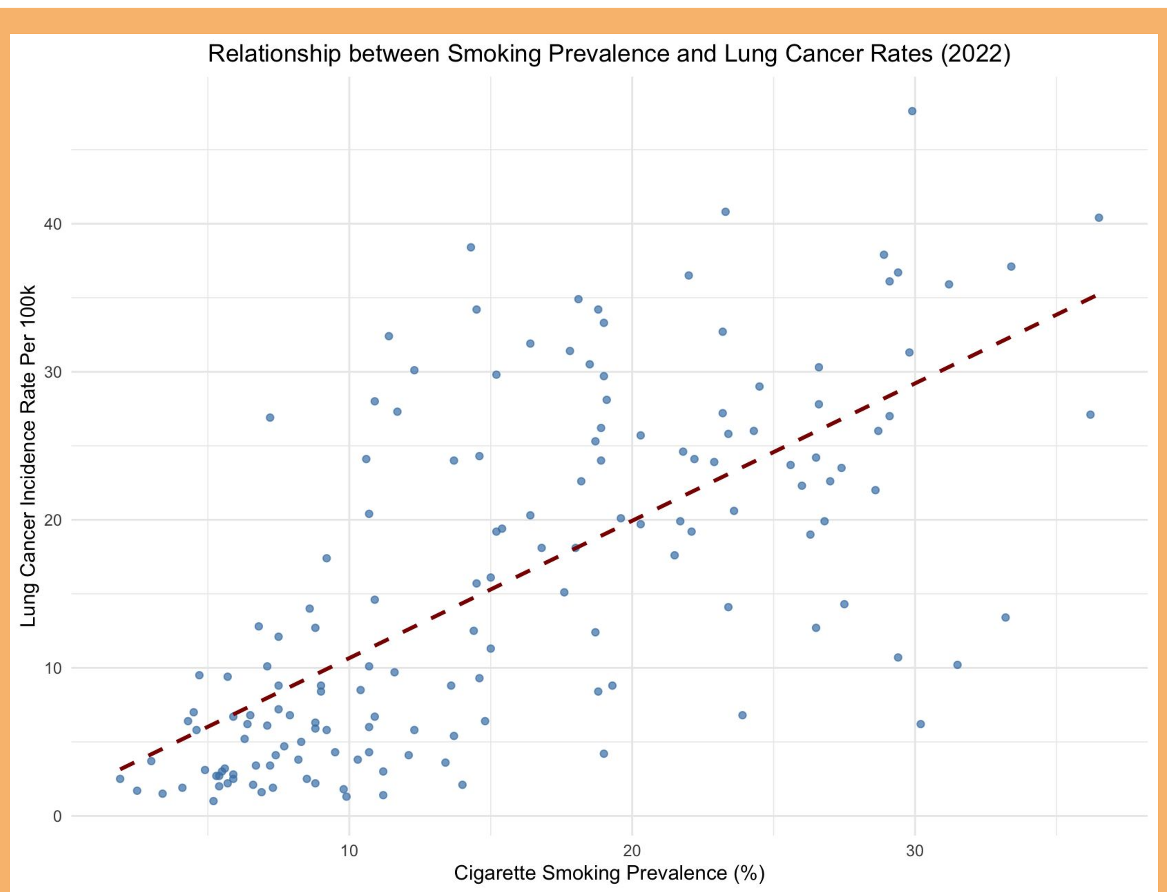

Figure 1: The Correlation between Tobacco Prevalence and Lung Cancer rates in 2022

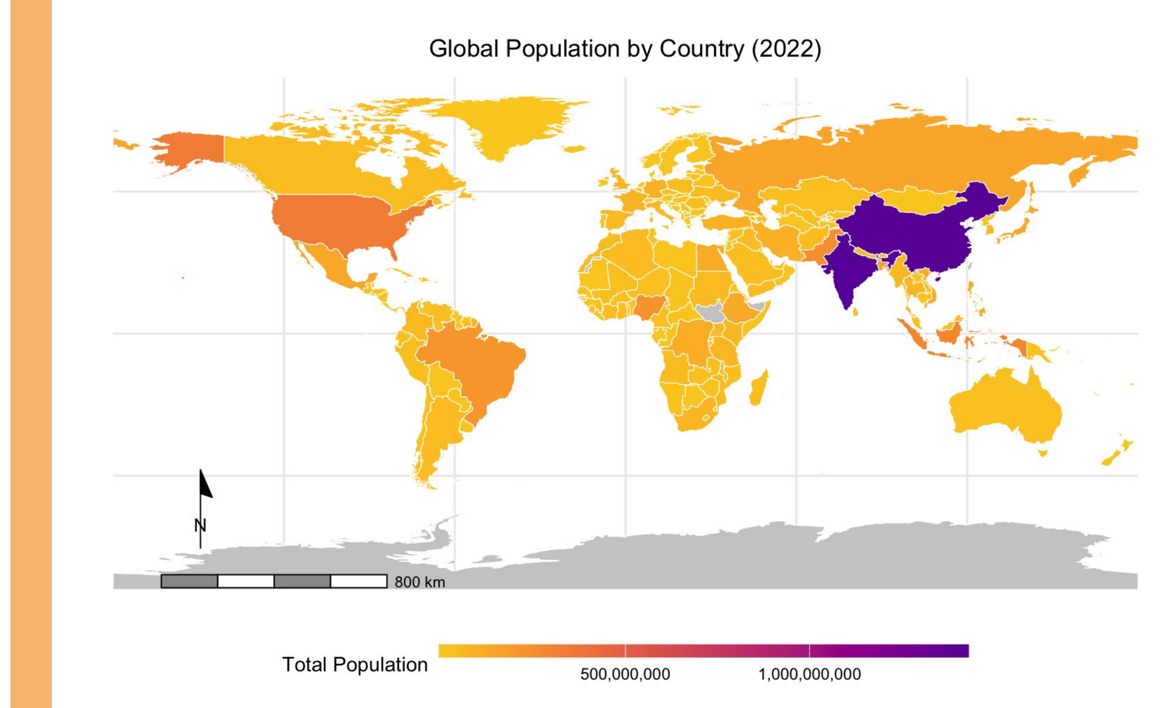

Figure 2: Global Population GIS Map 2022

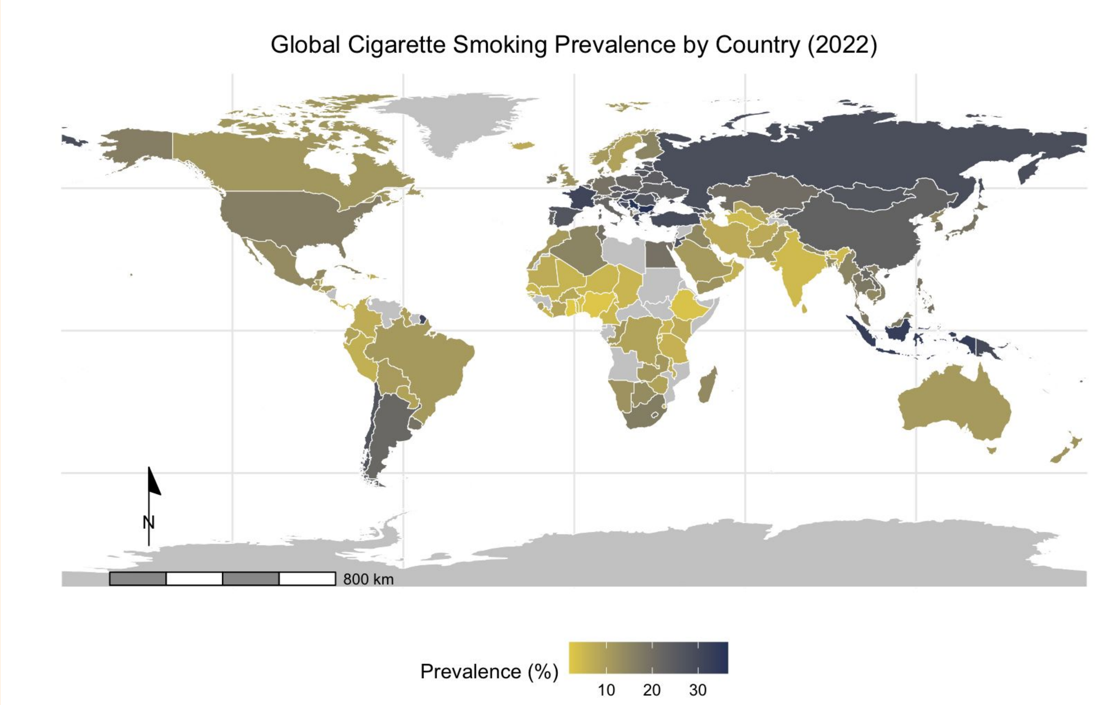

Figure 3: Global Cigarette Smoking Prevalence Rates GIS Map 2022

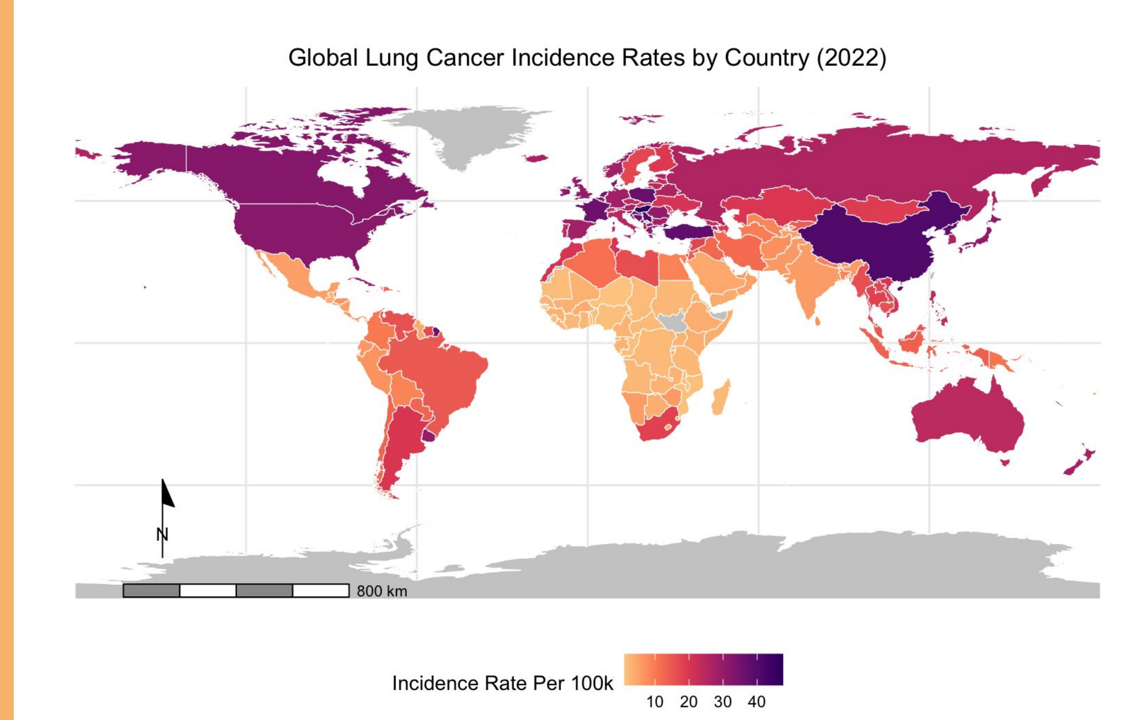

Figure 4: Global Lung Cancer Incidence Rates GIS Map 2022

Discussion/Conclusion:

- Many countries lack complete datasets for tobacco prevalence and lung cancer rates, possibly due to difficulties in retrieving data.
- Lung cancer generally increases with higher population density and tobacco prevalence.
- Some countries deviate from this trend, often due to variations in laws and enforcement.
- Tobacco prevalence affects lung cancer rates not only through direct smoking but also via secondhand smoke.
- The datasets cover only 2022, meaning population density, tobacco prevalence, and lung cancer rates may have changed since then.
- Studies could focus on trends in tobacco prevalence over time and their impact on lung cancer incidence.

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